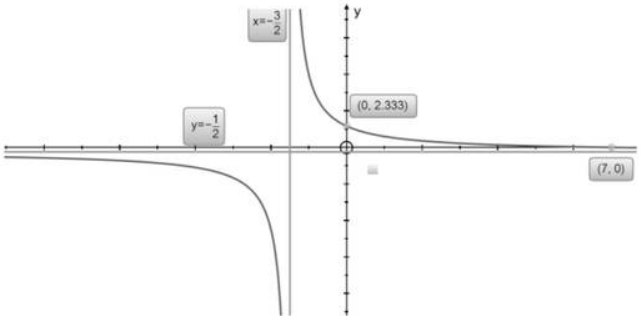
Type MS · Finding where a curve has a given gradient

Answer 1 of 10 · Question 10 part e · 7 marks

2025 JUN · Paper 02 · extract 1 of 5 · find part e in this mark scheme extract

10 (a)	(i) $x=-\frac{3}{2}$ (ii) $y=-\frac{1}{2}$	B1 B1 [2]
(b)	$(7,0)$ $\left(0,\frac{7}{3}\right)$	B1 B1 [2]
(c)		B1ft curve B1ft asymptotes B1ft (intersections with x- and y-axes) [3]
(d)	$\frac{-1(2x+3)-2(7-x)}{(2x+3)^2}=\left[\frac{-17}{(2x+3)^2}\right]$	M1
	$\frac{-1\left(2\times\frac{1}{2}+3\right)-2\left(7-\times\frac{1}{2}\right)}{\left(2\times\frac{1}{2}+3\right)^2}=-\frac{17}{16}$	M1A1* cso [3]
	ALT – product rule	
	$-1(2x+3)^{-1}+(7-x)(-1)(2)(2x+3)^{-2}=\left[\frac{-17}{(2x+3)^2}\right]$	M1
	$-1\left(2\times\frac{1}{2}+3\right)^{-1}+\left(7-\frac{1}{2}\right)(-1)(2)\left(2\times\frac{1}{2}+3\right)^{-2}=-\frac{17}{16}$	M1A1* cso

Type MS · Finding where a curve has a given gradient

Answer 1 of 10 · Question 10 part e · 7 marks

2025 JUN · Paper 02 · extract 2 of 5 · find part e in this mark scheme extract

(e)	$-\frac{17}{(2x+3)^2}=-\frac{17}{16}$	M1
	$(2x+3)^2=16\Rightarrow 2x+3=\pm\sqrt{16}\Rightarrow x=0.5 \text{ or } -3.5$	dM1
	$y=\frac{7--3.5}{2\times-3.5+3}\left(=-\frac{21}{8}\right)$	ddM1
	$\left(-\frac{7}{2},-\frac{21}{8}\right)$	A1
	$y--\frac{21}{8}=-\frac{17}{16}\left(x--\frac{7}{2}\right)$	M1A1
	$34x+32y+203=0$	A1 [7]
		Total 17 marks

Type MS · Finding where a curve has a given gradient

Answer 1 of 10 · Question 10 part e · 7 marks

2025 JUN · Paper 02 · extract 3 of 5 · find part e in this mark scheme extract

Part	Mark	Additional Guidance
For parts (a) and (b) look for their responses in the body of the question. In part (a), if the responses are not labelled (i) and (ii) accept them in order.		
(a)	(i) B1	For $x=-\frac{3}{2}$ oe
	(ii) B1	For $y=-\frac{1}{2}$ oe
(b)	B1 B1	First B1 for either correct, second B1 for both correct Condone if not given as coordinates e.g. $x=7$ and/or $y=\frac{7}{3}$ given
(c)	B1ft	Two branches drawn in the correct two "quadrants" created by the two asymptotes. Mark intention, allow poor curves, but do not allow the curve to bend back on itself or touch any asymptotes.
	B1ft	Two clearly marked asymptotes, ft their (a), labelled as described, there must be one section of the curve present, tending towards these asymptotes.
	B1ft	Two clearly labelled intersections with the axes, ft their (b), at least one section of their curve must pass through one of these intersections. Intersections must be labelled correct way around. If additional intersections seen then B0
(d)	M1	Attempt the quotient rule. Numerator must be of the form $A.(2x+3)-B.(7-x)$, $B>1$. Denominator must be of the form $(2x+3)^2$
	M1	Substitutes $x=\frac{1}{2}$ into their derivative **NB: This is an A mark in Epen**
	A1* cso	Obtains the required value of gradient $\left(-\frac{17}{16}\right)$ with no errors **NB: This is a B mark in Epen**
	ALT – product rule	
	M1	For an attempt at Product Rule. Must be a sum of two products. Must have the form $c(2x+3)^{-1}+d(7-x)(2x+3)^{-2}$ for constants c, d.
	M1	Substitutes $x=\frac{1}{2}$ into their derivative **NB: This is an A mark in Epen**
	A1* cso	Obtains the required value of gradient $\left(-\frac{17}{16}\right)$ with no errors **NB: This is a B mark in Epen**
(e)	M1	For equating the gradient of the tangent to their $\frac{dy}{dx}$ from (d)
	dM1	For rearranging to obtain $(2x+3)^2=16$ and attempt to solve or for rearrangement to a 3TQ and method to solve the 3TQ. See general guidance for acceptable methods to solve.

Type MS · Finding where a curve has a given gradient

Answer 1 of 10 · Question 10 part e · 7 marks

2025 JUN · Paper 02 · extract 4 of 5 · find part e in this mark scheme extract

	ddM1	For substitution of their x into the original equation to obtain y. This must **not** be $x=\frac{1}{2}$ This M mark is dependent on the **two** previous M marks
	A1	For correct coordinate of $P\left(-\frac{7}{2},-\frac{21}{8}\right)$
	M1	For substitution of their coordinates of P and the gradient $-\frac{17}{16}$ to obtain an equation of a straight line using any valid method. If $y=mx+c$ is used a value of c must be found **and** an equation needs to be formed. For example; $'-\frac{21}{8}'=-\frac{17}{16}\times'-\frac{7}{2}'+c\Rightarrow c=-\frac{203}{32}\Rightarrow y=-\frac{17}{16}x-\frac{203}{32}$ [
	A1	For any correct equation of the tangent with the correct values of x and y in any form.
	A1	For the correct equation of tangent given in the required form. $34x+32y+203=0$

Type MS · Finding where a curve has a given gradient

Answer 1 of 10 · Question 10 part e · 7 marks

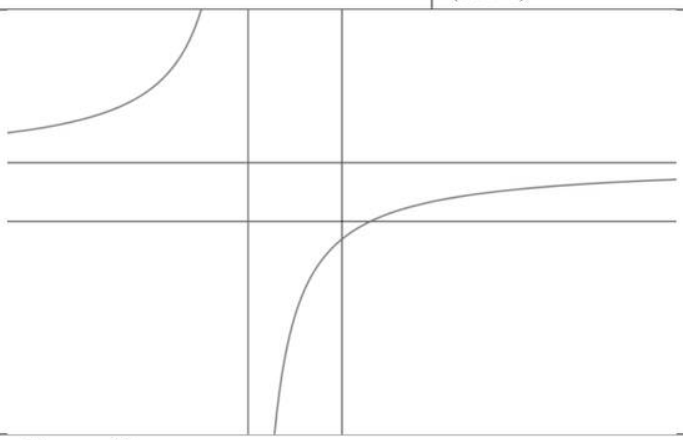
2025 JUN · Paper 02 · extract 5 of 5 · find part e in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding where a curve has a given gradient

Answer 2 of 10 · Question 10 part f · 6 marks

2025 JUN · Paper 02 (regional R) · extract 1 of 4 · find part f in this mark scheme extract

10(a) Mark (i) and (ii) together	$-\frac{1}{4}=-\frac{2}{b}\rightarrow b=8$ * $-\frac{4}{3}a+8=0$ oe or $-\frac{b}{a}=-\frac{4}{3}$ $a=6$	B1*cso M1 A1 [3]
(b)	$y=\frac{5}{"6"}$	B1ft [1]
(c)	$\left(\frac{2}{5},0\right)$	B1 [1]
(d)	$\frac{5(6x+8)-6(5x-2)}{(6x+8)^2}$ $\frac{52}{(6x+8)^2}$ oe eg $\frac{52}{36x^2+96x+64}$ $\frac{52}{(6x+8)^2}$ and correct conclusion ALT using product rule $(5x-2)(6x+8)^{-1}$ $5(6x+8)^{-1}+(-1)(6)(5x-2)(6x+8)^{-2}$ $\frac{52}{(6x+8)^2}$ oe eg $\frac{52}{36x^2+96x+64}$ $\frac{52}{(6x+8)^2}$ and correct conclusion	M1 dM1 A1 A1 [4]
(e)		B1 (curve) B1ft (asymptotes) B1ft (intersections with axes) [3]
(f)	$\frac{13}{100}=\frac{52}{(6x+8)^2}$ oe $(6x+8)^2=400$ **$3x^2+8x-28=0$ oe** $x=2, y=\frac{2}{5}$ $x=-\frac{14}{3}, y=\frac{19}{15}$ oe $\sqrt{\left(("2")-\left("-\frac{14}{3}"\right)\right)^2+\left(\left("\frac{2}{5}"\right)-\left("\frac{19}{15}"\right)\right)^2}$ 6.72	M1 dM1 A1 A1 dM1 A1 [6]
		Total 18 marks

Type MS · Finding where a curve has a given gradient

Answer 2 of 10 · Question 10 part f · 6 marks

2025 JUN · Paper 02 (regional R) · extract 2 of 4 · find part f in this mark scheme extract

Part	Mark	Additional Guidance
(a) (i) and (ii)	B1*cso	For a minimum of the steps shown in the MS to find b, no errors.
	M1	For correct substitution of $x=-\frac{4}{3}$ and $b=8$ into $ax+b=0$
	A1	Correct a.
		Students can complete this in a different order $-\frac{4}{3}a+8=0 \Rightarrow a=6$ $-\frac{b}{a}=-\frac{4}{3}\left(\text{or } -\frac{b}{6}=-\frac{4}{3}\right)\Rightarrow b=8$ B1 M1 A1
(b)	B1ft	Follow through their value of a. $y=\frac{5}{"their 6"}$
(c)	B1	$\left(\frac{2}{5},0\right)$ allow $x=\frac{2}{5}$ $y=0$ $x=\frac{2}{5}$ alone is insufficient
(d)	M1	Attempt the quotient rule. Numerator must be the difference of two terms of the form $A\times(ax+8)-B\times(5x-2)$ or $B\times(5x-2)-A\times(ax+8)$, A and $B>1$. Denominator must be of the form $(ax+8)^2$ $a\neq 0$ Allow follow through of their value of a (general principle of method mark).
	dM1	Either term on the numerator fully correct (either way round), dependent on previous method mark, doesn't need to be simplified. Denominator must be of the form $(ax+8)^2$ Allow follow through of their value of a (general principle of method mark).
	A1	$\frac{52}{(6x+8)^2}$ oe eg $\frac{52}{36x^2+96x+64}$ eg $\frac{13}{(3x+4)^2}$ A simplified expression (denominator expanded acceptable) Dependent on both previous method marks – question states must use calculus, so some method must be shown.
	A1	$\frac{52}{(6x+8)^2}$ and correct conclusion, for example, (the numerator is a positive number) eg the denominator is always positive eg the $(ax+8)^2>0$ and therefore (the fraction) is always positive. No incorrect work for this mark eg don't accept incorrectly simplified gradient.
ALT	M1	Attempt the product rule. $C\times(ax+8)^{-1}+(5x-2)(-(1))(D)\ (ax+8)^{-2}$, C and $D>1$. Allow follow through of their value of a (general principle of method mark).
	M1	Either term correct (either way round), dependent on previous method mark, doesn't need to be simplified
	A1 A1	As main scheme

Type MS · Finding where a curve has a given gradient

Answer 2 of 10 · Question 10 part f · 6 marks

2025 JUN · Paper 02 (regional R) · extract 3 of 4 · find part f in this mark scheme extract

(e)	B1	For a negative reciprocal curve drawn anywhere in the grid – there **must** be two branches present, they **must not** cross any asymptotes drawn or implied or cross each other and **must not obviously** 'bend back' on themselves. Mark intention.
	B1ft	Two clearly marked asymptotes $x = -\frac{8}{\text{their } a}$, $y = \frac{5}{\text{their } a}$, ft their value for a, labelled as equations or clearly passing through the correctly labelled numbers on the axis. There must be one section of a negative reciprocal curve present, tending towards these asymptotes. This branch must not obviously cross or bend back from any asymptotes drawn or implied Ignore any other branch/curve/line drawn if there is one branch fulfilling these conditions.
	B1ft	Single curve (or even line) passing through $\left(0, -\frac{1}{4}\right)$ and $\left(\frac{2}{5}, 0\right)$ marked clearly on the graph as coordinates or crossing points on the axis. Ignore any other branch/curve/line present. Allow follow through only of their $(c, 0)$ $c \neq 0$ from part (c)
(f)	M1	Sets their differentiated function from part (d) = $\frac{13}{100}$ Allow any function = $\frac{13}{100}$ coming from working in part (d) or re-started work to find the derivative. The function may not be the original function.
	dM1	For correctly rearranging an equation of the form $\frac{13}{100} = \frac{Q}{(cx+d)^2}$ to $(cx+d)^2 = R$ using their a, or to the form $px^2 + qx + r = 0$ $p, q, r \neq 0$ with 2 out of 3 of p, q, r correct. leading to $x =$
	A1	For $x = 2, y = \frac{2}{5}$ or $x = -\frac{14}{3}, y = \frac{19}{15}$ The pairing must be clear (may be implied in further working)
	A1	For $x = 2, y = \frac{2}{5}$ and $x = -\frac{14}{3}, y = \frac{19}{15}$ The pairing must be clear (may be implied in further working)
	dM1	For correct use of the formula to find the length of a line, using their values for x and y. Dependent on **first** method mark only.
	A1	For awrt 6.72

Type MS · Finding where a curve has a given gradient

Answer 2 of 10 · Question 10 part f · 6 marks

2025 JUN · Paper 02 (regional R) · extract 4 of 4 · find part f in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding where a curve has a given gradient

Answer 3 of 10 · Question 10 part d.i · 0 marks

2022 JAN · Paper 02 · find part d.i in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Finding where a curve has a given gradient

Answer 4 of 10 · Question 10 part d.ii · 8 marks

2022 JAN · Paper 02 · find part d.ii in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Finding where a curve has a given gradient

Answer 5 of 10 · Question 10 part e.i · 0 marks

2022 JAN · Paper 02 · find part e.i in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Finding where a curve has a given gradient

Answer 6 of 10 · Question 10 part e.ii · 3 marks

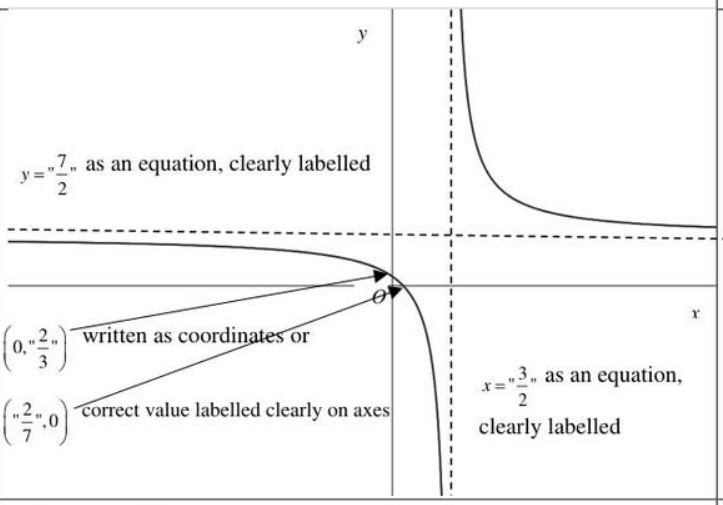
2022 JAN · Paper 02 · find part e.ii in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Finding where a curve has a given gradient

Answer 7 of 10 · Question 10 part e · 7 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 5 · find part e in this mark scheme extract

10 (a) (i)	$x = \frac{3}{2}$	B1
(ii)	$y = \frac{7}{2}$	B1 [2]
(b)	$\left(\frac{2}{7}, 0\right)$ $\left(0, \frac{2}{3}\right)$	B1 B1 [2]
(c)	$\frac{7(2x-3)-2(7x-2)}{(2x-3)^2}$	M1 A1
	$\frac{-17}{(2x-3)^2}$ or $\frac{-17}{4x^2-12x+9}$	A1
	Correct conclusion	B1 [4]
	ALT – product rule	M1 A1
	$7(2x-3)^{-1} + (7x-2)(-1)(2)(2x-3)^{-2}$	
	$\frac{-17}{(2x-3)^2}$ or $\frac{-17}{4x^2-12x+9}$	A1
	Correct conclusion	B1
(d)		B1 (curve) B1ft (asymptotes) B1ft (intersections with x- and y-axes) [3]
(e)	$-\frac{1}{17} = "\frac{-17}{(2x-3)^2}"$	M1
	"$(2x-3)^2 = 17^2$" **or** "$4x^2 - 12x - 280 = 0$" oe	dM1
	$x = 10$ $y = 4$ or $(10, 4)$	A1
	$y - "4" = 17(x - "10")$ **or** "4" = –17 × "10" + c leading to $c =$	M1
	$y = 17x - 166$ oe	A1

Type MS · Finding where a curve has a given gradient

Answer 7 of 10 · Question 10 part e · 7 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 5 · find part e in this mark scheme extract

	"$17x - 166" = \dfrac{7x-2}{2x-3}$ → $17x^2 - 195x + 250$	M1
	$x = \dfrac{25}{17}$ $y = -141$ or $\left(\frac{25}{17}, -141\right)$	A1 [7]
		Total 18 marks

Type MS · Finding where a curve has a given gradient

Answer 7 of 10 · Question 10 part e · 7 marks

2022 JUN · Paper 02 (regional R) · extract 3 of 5 · find part e in this mark scheme extract

Part	Mark	Additional Guidance
		If a candidate gives **no response** to (a) and/or (b) but shows the correct answers on the graph we will award the marks. Where answers are given in (a) and/or (b) these should be marked as they stand with no reference to the graph. Ignore labelling of (i) and (ii) and mark (a) together.
(a)(i)	B1	For $x = \frac{3}{2}$ oe
(a)(ii)	B1	For $y = \frac{7}{2}$ oe
(b)	B1 B1	First B1 for either correct, second B1 for both correct Condone if not given as coordinates e.g. $x = \frac{2}{7}$ and/or $y = \frac{2}{3}$ given
(c)	M1	Attempt the quotient rule. Numerator must be the difference of two terms (either way round) of the form $A.(2x-3) - B.(7x-2)$, A and $B > 1$. Denominator must be of the form $(2x-3)^2$
	A1	Either term on the numerator correct (either way round), dependent on previous method mark.
	A1	Obtains $\frac{-17}{(2x-3)^2}$ or $\frac{-17}{4x^2-12x+9}$
	B1	Correct conclusion based on correct working only, for example, (the numerator is a negative number and) the denominator is always positive and therefore the fraction/gradient is always negative.
	ALT – product rule	
	M1	For an attempt at Product Rule. Must be a sum of two products. Must have the form $c(2x-3)^{-1} + d(7x-2)(2x-3)^{-2}$ for constants c, d.
	A1	Either term correct, dependent on previous method mark.
	A1	Obtains $\frac{-17}{(2x-3)^2}$ or $\frac{-17}{4x^2-12x+9}$
	B1	Correct conclusion based on correct working only, for example, (the numerator is a negative number and) the denominator is always positive and therefore the fraction/gradient is always negative.
(d)	B1	Two branches drawn in the correct two "quadrants" created by the two aymptotes. Mark intention, allow poor curves, but do not allow the curve to bend back on itself or touch any asymptotes. Allow BOD if intention is for curve to run alongside asymptote but there is a slight deviation back on itself.
	B1ft	Two clearly marked asymptotes, ft their (a), labelled as described, there must be one section of the curve present, tending towards these asymptotes.
	B1ft	Two clearly labelled intersections with the axes, ft their (b), at least one section of their curve must pass through one of these intersections. Intersections must be labelled correct way around. If additional intersections seen then B0
(e)	M1	Sets their differentiated function from part (c) = $-\frac{1}{17}$
	dM1	Rearranges to get to an equation of the form shown with no denominators or a 3TQ and solves using an acceptable method to obtain $x = \ldots$. Dependent on previous method mark
	A1	Correct values for point A (10, 4)
	M1	Uses their values for x and y (from an attempt at working with gradient of the curve) with gradient 17 to find an equation for l (if using $y = mx + c$, must be a complete method arriving at $c =$) If correct $c = -166$.
	A1	Correct equation, any form

Type MS · Finding where a curve has a given gradient

Answer 7 of 10 · Question 10 part e · 7 marks

2022 JUN · Paper 02 (regional R) · extract 4 of 5 · find part e in this mark scheme extract

	M1	Sets their equation for the normal equal to the curve, makes a **correct** rearrangement to remove any denominator and forms a 3TQ Note this method mark is not dependant.
	A1	Correct exact values for x and y

Type MS · Finding where a curve has a given gradient

Answer 7 of 10 · Question 10 part e · 7 marks

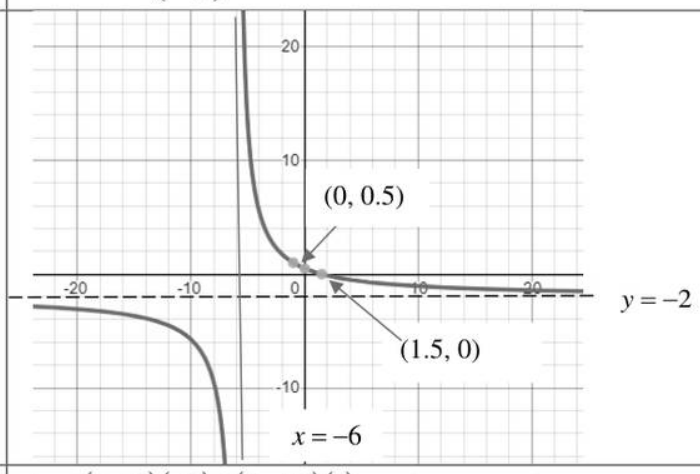
2022 JUN · Paper 02 (regional R) · extract 5 of 5 · find part e in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding where a curve has a given gradient

Answer 8 of 10 · Question 9 part e · 5 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 4 · find part e in this mark scheme extract

9(a)	(i) $y=-2$ (ii) $x=-6$	B1 B1 [2]
(b)	(i) $\left(\frac{3}{2},0\right)$ (ii) $\left(0,\frac{1}{2}\right)$	B1 B1 [2]
(c)		B1 – shape B1ft– Asymptotes B1ft – Intersections [3]
(d)	$\frac{dy}{dx}=\frac{(x+6)(-2)-(3-2x)(1)}{(x+6)^2}$ $\frac{dy}{dx}=\frac{-15}{(x+6)^2}$ with conclusion; numerator negative, denominator always positive, $\frac{-}{+}\Rightarrow$ negative	M1A1 A1 [3]
(e)	$\frac{-15}{(x+6)^2}=-\frac{3}{5}\Rightarrow 25=(x+6)^2\Rightarrow x=-6\pm 5=-11,\quad -1$ $y=\frac{3-2(-1)}{-1+6}=1$ $y-1=-\frac{3}{5}(x-[-1])\Rightarrow y=-\frac{3}{5}x+\frac{2}{5}\Rightarrow k=\frac{2}{5}$	M1A1 B1 M1A1 [5]
		Total 15 marks

Type MS · Finding where a curve has a given gradient

Answer 8 of 10 · Question 9 part e · 5 marks

2023 JUN · Paper 02 (regional R) · extract 2 of 4 · find part e in this mark scheme extract

Part	Mark	Notes
(a) (i)	B1	For the correct equation. If their equations are not labelled (i) and (ii) accept them in the order given only. For example, the following presentation is B0B0 $x=-6$ $y=-2$
(ii)	B1	For the correct equation.
(b)		If their coordinates are not labelled (i) and (ii) or are given in the incorrect place [for example, (b)(i) $y=\frac{1}{2}$ is B0] accept them in the order given only. For example, the following presentation is B0B0 $\left(0,\frac{1}{2}\right)$ $\left(\frac{3}{2},0\right)$
(i)	B1	For the correct coordinates. Accept $x=\frac{3}{2}$
(ii)	B1	For the correct coordinates. Accept $y=\frac{1}{2}$
(c)	B1	For the correct shape with two branches the correct way around anywhere in the grid. It must be asymptotic in nature. The ends of the curve must not come back on themselves. Whilst you need to be fairly generous, any obvious turning back is B0
	B1ft	For the correct asymptotes drawn and labelled with at least one branch of the curve [which must be asymptotic in nature in at least one branch] in the correct place seen. Ft their asymptotes Accept the vertical line drawn shown passing through their -6 and the horizontal line drawn shown passing through their -2
	B1ft	For the correct coordinates of intersections with the relevant branch of the curve drawn correctly seen. The curve must go through the points. Do not accept touching the axis. Allow $\frac{1}{2}$ marked on y-axis and $\frac{3}{2}$ marked on the x-axis. Ft their coordinates of intersections.
(d)	M1	For an attempt at quotient rule. • The denominator must be squared, or accept $(x+6)(x+6)$ or $x^2+12x+36$ • Both $(x+6)$ and $(3-2x)$ differentiated correctly. • The two terms in the numerator subtracted either way around.
	A1	Fully correct derivative (simplification not required for this mark).
	A1	For a correct simplified derivative with a correct conclusion. For example: $(x+6)^2 \geqslant 0$, -15 is negative, $\frac{\text{negative}}{\text{positive}}$ is always negative. [Accept also $(x+6)^2>0$]

Type MS · Finding where a curve has a given gradient

Answer 8 of 10 · Question 9 part e · 5 marks

2023 JUN · Paper 02 (regional R) · extract 3 of 4 · find part e in this mark scheme extract

(e)	M1	For setting the value of $-\frac{3}{5}$ = their $\frac{dy}{dx}$ with an attempt to find at least one value of x. Allow this even if their derivative results in a linear equation.
	A1	For **both** correct values of x
	B1	For $y=1$ using $x=-1$ OR For $y=-5$ using $x=-11$
	M1	For forming an equation of the line with either $x=-1,\quad y=1$ or $x=-11,\quad y=-5$ or their x and their y
	A1	For the correct value of k (accept an embedded value). $y=-\frac{3}{5}x+\frac{2}{5}\Rightarrow k=\frac{2}{5}$ You can award this mark even if the previous A mark has not been scored. So, for a correct solution without showing that $x=-11$ score M1A0B1M1A1
	ALT for last 3 marks	
	B1	Sets $-\frac{3}{5}x+k=\frac{3-2x}{x+6}$
	M1	Substitutes $x=-1$ or -11 into the above equation
	A1	For $k=\frac{2}{5}$ You can award this mark even if the previous A mark has not been scored. So, for a correct solution without showing that $x=-11$ score M1A0B1M1A1

Type MS · Finding where a curve has a given gradient

Answer 8 of 10 · Question 9 part e · 5 marks

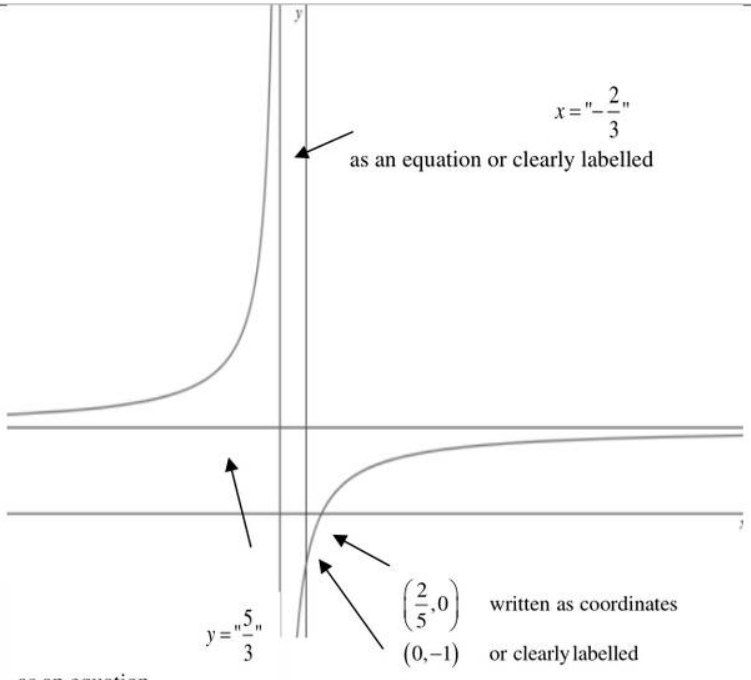
2023 JUN · Paper 02 (regional R) · extract 4 of 4 · find part e in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding where a curve has a given gradient

Answer 9 of 10 · Question 10 part d · 11 marks

2024 NOV · Paper 02 · extract 1 of 3 · find part d in this mark scheme extract

10 (a)	(i) $\left(\frac{2}{5},0\right)$ (ii) $(0,-1)$	B1 B1 [2]
(b) (i) (ii)	$y=\frac{5}{3}$ $x=-\frac{2}{3}$	B1 B1 [2]
(c)	 as an equation or clearly labelled	B1 B1ft B1ft [3]
(d)	$\left\{y=\frac{1}{4}x+\frac{7}{4}\right\}\Rightarrow\frac{1}{4}$	B1
	$\left\{\frac{dy}{dx}=\right\}\frac{5(3x+2)-3(5x-2)}{(3x+2)^2}\left\{=\frac{16}{(3x+2)^2}\right\}$	M1 dM1A1
	$"\frac{1}{4}"="\frac{16}{(3x+2)^2}"\Rightarrow x=...$	M1
	$x=2$ and $y\left\{=\frac{5\times"2"-2}{3\times"2"+2}\right\}=1$	A1
	$y-"1"="-4"(x-"2")$	M1 dM1
	$(0,9)$ and $\left(\frac{9}{4},0\right)$	A1ft
	$\left\{\lvert DE\rvert\right\}=\sqrt{9^2+\left(\frac{9}{4}\right)^2}=\frac{9\sqrt{17}}{4}$	M1A1
		[11]
		Total 18 marks

Type MS · Finding where a curve has a given gradient

Answer 9 of 10 · Question 10 part d · 11 marks

2024 NOV · Paper 02 · extract 2 of 3 · find part d in this mark scheme extract

Part	Mark	Additional Guidance
(a)(i)	B1	$\left(\frac{2}{5},0\right)$ Accept $x=\frac{2}{5},\{y=0\}$
(a)(ii)	B1	$(0,-1)$ Accept $\{x=0\}, y=-1$, if not labelled, we score by their order
(b)(i)	B1	Correct equation $y=\frac{5}{3}$
(b)(ii)	B1	Correct equation $x=-\frac{2}{3}$
(c)	B1	Two branches drawn in the correct two "quadrants" created by the two asymptotes. Mark intention, allow poor curves, but do not allow the curve to bend back on itself or touch any asymptotes.
	B1ft	Two clearly labelled intersections with the axes, ft their (a), at least one section of their curve must pass through one of these intersections.
	B1ft	Two clearly marked asymptotes, ft their (b), labelled as described, there must be at least one section of the curve present, not crossing the curve.
(d)	B1	$\frac{1}{4}$ seen any where in part (d)
	M1	Attempt to differentiate y **Quotient rule**: Numerator must be the difference of two terms (either way round) of the form $P(3x+2)-Q(5x-2),\quad P,Q>0$. Denominator must be correct. **Product rule**: obtains form $P(3x+2)^{-1}-Q(5x-2)(3x+2)^{-2},\ P,Q>0$
	dM1	**Quotient rule**: Either term in the numerator correct **Product rule**: Either term correct Dependent on previous method mark.
	A1	Correct $\frac{dy}{dx}$. May be unsimplified.
	M1	Sets their $\frac{1}{4}$ equal to their $\frac{dy}{dx}$ and attempts to solve for x
	A1	$x=2$ *and* $y=1$
	M1	Finds the gradient of the normal using $\frac{-1}{their\ \frac{1}{4}}$
	dM1	A complete and correct method to find the equation of the normal. If using $y=mx+c$, c must be found. Dependent on previous method mark.
	A1ft	$(0,9)$ and $\left(\frac{9}{4},0\right)$, accept $y=9,\ x=\frac{9}{4}$ Ft their normal equation.
	M1	Correct use of the formula for the length of a line segment, using their values from their points of D and E, may not come from the normal.
	A1	Correct exact length in simplest form

Type MS · Finding where a curve has a given gradient

Answer 9 of 10 · Question 10 part d · 11 marks

2024 NOV · Paper 02 · extract 3 of 3 · find part d in this mark scheme extract

Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom

Type MS · Finding where a curve has a given gradient

Answer 10 of 10 · Question 8 part a · 5 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

8 (a)	$\left(\frac{dy}{dx}=\right)4x-4$	M1
	Gradient of $l=8$	B1
	Gradient of normal $=-\frac{1}{8}$	B1
	$4x-4=-\frac{1}{8}\Rightarrow x=\frac{31}{32}$	M1A1 [5]
(b) (i)	$(\alpha+\beta)^3=\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3=\alpha^3+\beta^3+3\alpha\beta(\alpha+\beta)$	B1*cso [1]
(ii)	$\alpha^4+\beta^4=\left(\alpha^2+\beta^2\right)^2-2\alpha^2\beta^2=\left((\alpha+\beta)^2-2\alpha\beta\right)^2-2(\alpha\beta)^2$	B1*cso [1]
(c)	$\alpha+\beta=2\quad\alpha\beta=\frac{9}{2}$ oe	B1
	$\left(\alpha^3-\beta+\beta^3-\alpha=\alpha^3+\beta^3-(\alpha+\beta)\right)$	
	$=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)-(\alpha+\beta)$	M1
	(sum of roots $=$) $2^3-3\left(\frac{9}{2}\right)(2)-2(=-21)$	A1
	$\left[\left(\alpha^3-\beta\right)\left(\beta^3-\alpha\right)=\right]\alpha^3\beta^3-\alpha^4-\beta^4+\alpha\beta$	M1
	$=(\alpha\beta)^3-\left(\alpha^4+\beta^4\right)+\alpha\beta$	
	Now, $\left(\alpha^4+\beta^4\right)=\left((2)^2-2\times\frac{9}{2}\right)^2-2\times\left(\frac{9}{2}\right)^2=\left(-\frac{31}{2}\right)$	A1ft
	(product of roots $=$) $=\left(\frac{9}{2}\right)^3-\left(-\frac{31}{2}\right)+\frac{9}{2}\left(=\frac{889}{8}\right)$	A1
	$x^2-"-21"x+"\frac{889}{8}"$	M1
	$8x^2+168x+889=0$	A1 [8]
		Total 15 marks

Type MS · Finding where a curve has a given gradient

Answer 10 of 10 · Question 8 part a · 5 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
8 (a)	M1	For a correct differentiation.
	B1	For 8, identified as the gradient or used to calculate gradient of normal
	B1	For $-\frac{1}{8}$ stated as gradient of normal or used in subsequent work
	M1	For correctly solving their $ax+b=-\frac{1}{8}$
	A1	For $\frac{31}{32}$
(b) (i)	B1*cso	Complete and full algebra to show the given identity
(ii)	B1*cso	Complete and full algebra to show the given identity
(c)	B1	Correct values for $\alpha+\beta$ **and** $\alpha\beta$. Award if values not seen explicitly but embedded in the sum/product calculations for the new equation.
	M1	**Sum:** Using the result from part (b) and correctly manipulating to prepare for substitution of $\alpha+\beta$ **and** $\alpha\beta$
	A1	For -21
	M1	**Product:** For the correct algebraic expansion shown.
	A1ft	Correct substitution of their values into $\alpha^4+\beta^4$ within the product of roots
	A1	For $\frac{889}{8}$
	M1	Use $x^2-(\text{their sum of their roots})x+\text{their product of roots}$ "= 0" may be missing.
	A1	Correct equation oe with integer coefficients.

Type MS · Finding where a curve has a given gradient

Answer 10 of 10 · Question 8 part a · 5 marks

2025 JUN · Paper 01 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks